IA SUMMER PROGRAM 2024

It takes two to tango

Testing the ‘merging paradigm’ in AGN-host galaxies

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Tango?

An Active Galactic Nucleus (AGN) consists of the central region of a galaxy, wherein we can observe a manifestation of gas accretion onto a supermassive black hole (SMBH).

According to the unification theory, all AGN are the same type of object and share the same evolutionary process. Type I simply refers to AGN whose nucleus is visible, while Type II refers to AGN whose nucleus and broad line region are obscured; this may be due either to the viewing angle or some intrinsic difference in structure.

Could the chaos that occurs during the process of galaxy mergers (a sort of celestial tango) alter the torus-like structure of the AGN, obscuring it and making it a Type II?

Sample

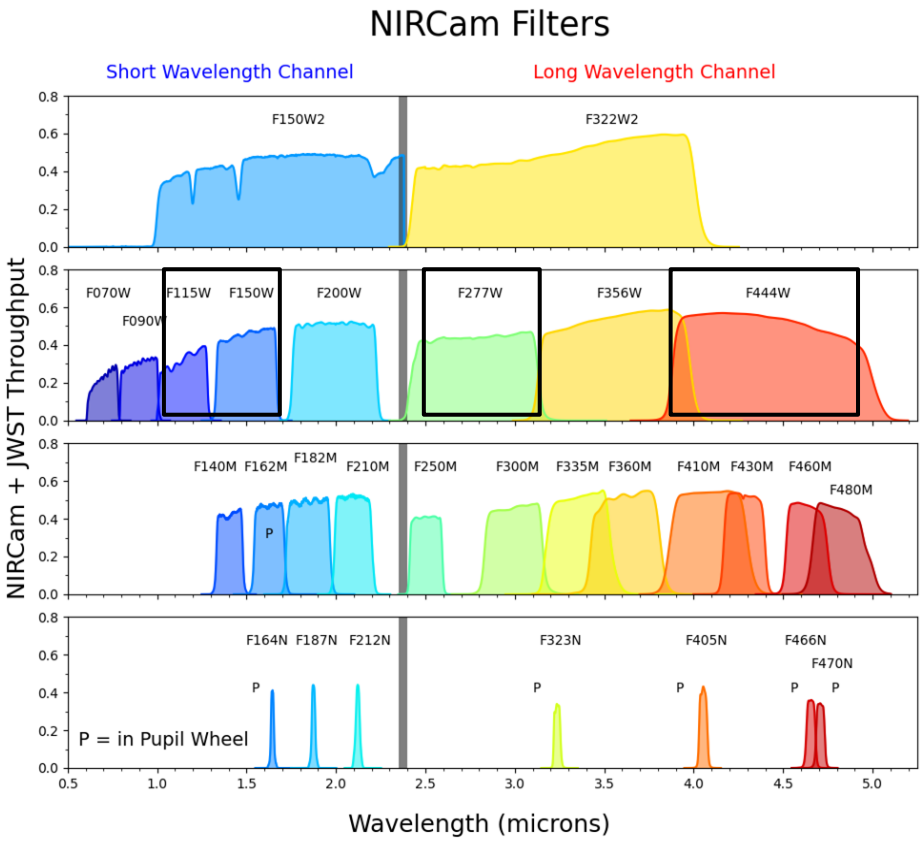
In this study, we used a sample of 60 galaxies with [NeV]3426 emission from Bongiorno et al. (2013), with redshift ∈ [0.6,1.2]. These galaxies are thought to be obscured AGNs, since this high-ionisation emission line is the only AGN-signature found in the optical spectrum.

The data used and analysed comes from the *Hubble Space Telescope* (*HST*) and *James Webb Space Telescope* (*JWST*).

For *HST* we have data for 60 galaxies from one filter (*f814w*). For *JWST* we have data for 24 galaxies from 4 different filters (*f115w, f150w, f277w, f444w*).

The filter *f814w* from *Hubble* is part of the *Wide Field Camera 3* (*WFC3*) and is used to observe galaxies in the near-infrared part of the spectrum.

The filters from *James Webb* are part of the *Near Infrared Camera* (*NIRCam*). The filters *f115w* and *f150w* are in the short wavelength channel and the filters *f227w* and *f444w* are in the long wavelength channel.



Parametric Methods

Fit predefined models to the galaxy's light profile based on physical assumptions.

Galfit: fits different light profiles (Sérsic, Nuker, Exponential Disk, ...) to galaxy images, using chi-squared minimization.

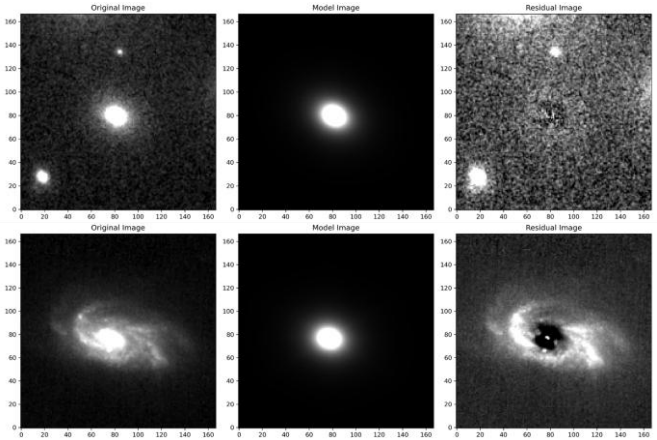
PySersic: fits the Sérsic profile to astronomical images, using Bayesian inference.

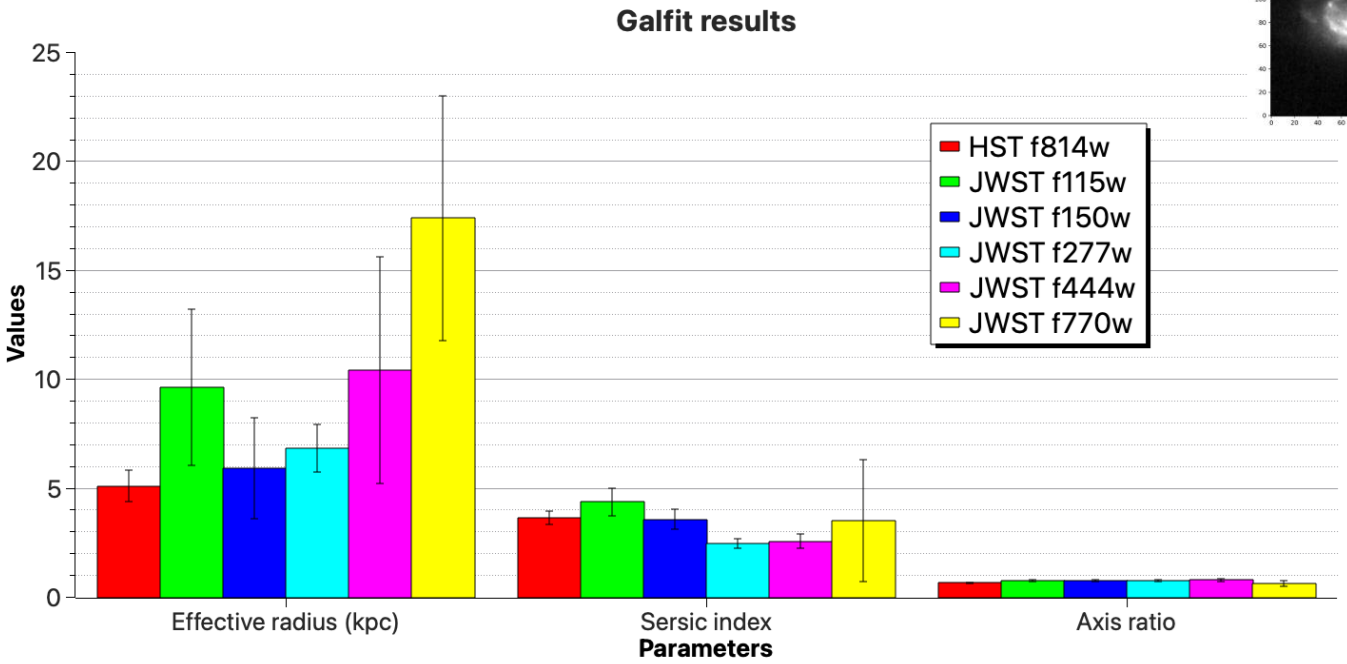
PyAutoGalaxy: fits different light profiles, including more complex ones (Elliptical Sérsic), to galaxy images, using probabilistic methods and Bayesian inference.

GALight: fits different light profiles in a more flexible and user-friendly way, also using chi-squared minimization; it can also be a non-parametric method.

Results Galfit

Downside: fitting is not very good for galaxies with arms





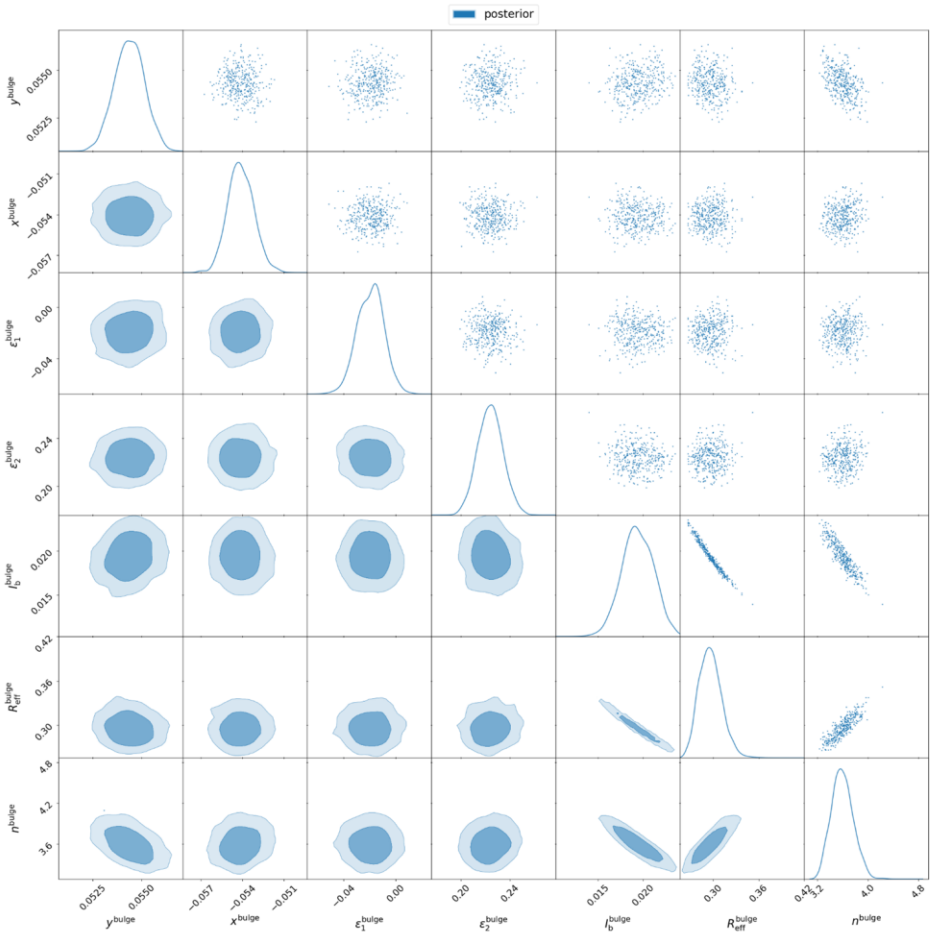
- **Effective radius:** within which half of the total light is emitted
- **Sérsic index:** the higher the more concentrated the galaxy is
- **Axis ratio:** semi-minor axis / semi-major axis

Results PyAutoGalaxy

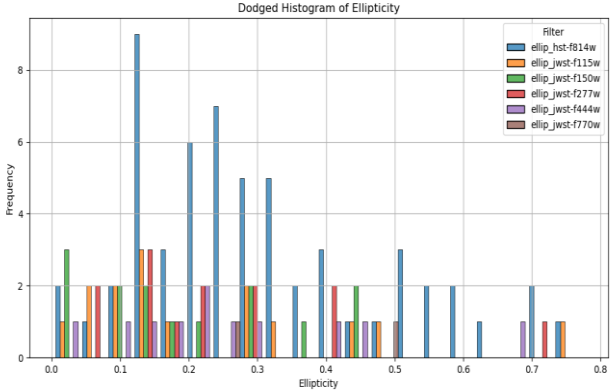
Hubble Space Telescope - filter *f814w*

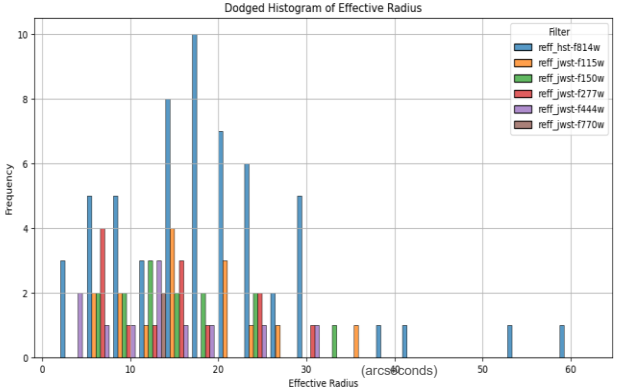
Median Effective Radius: 5,3601 kpc
Median Sersic Index: 4,4431
Median Magnitude: 0,3714

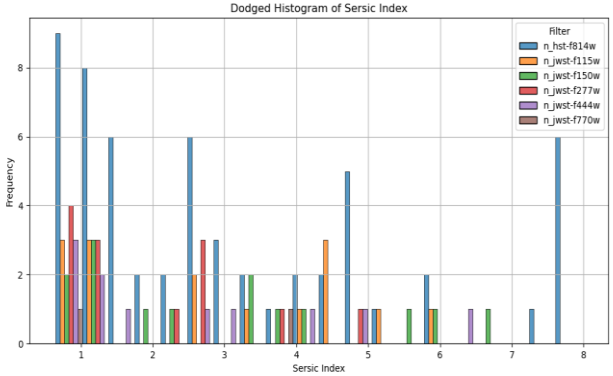
Downside: Can only utilize the CPU for computations. Inefficiency stems from the high computational demands of the complex algorithms used for modeling galaxies, which, when executed without parallel processing or GPU acceleration, leads to extended runtimes (± 1h20 per galaxy).



Results Pysersic







Downside: The package might not handle missing or incomplete data gracefully, requiring pre-processing steps to clean the data before analysis.

Non-Parametric Methods

Use statistical measures to describe galaxy morphology without assuming any specific model.

CAS

Concentration

$$C = 5\log_{10}\left(\frac{r_{80}}{r_{20}}\right)$$

Asymmetry

$$A = \frac{\sum|I - I_{180}|}{\sum|I|} - \frac{\sum|B - B_{180}|}{\sum|I|}$$

Clumpiness

$$S = \frac{\sum|I - I_S|}{\sum|I|} - \frac{\sum|B - B_S|}{\sum|I|}$$

Gini-M20

Gini Coefficient

$$\mathrm{G} = \frac{1}{|\bar{f}|n(n-1)}\sum_{i}^{n}(2i - n - 1)|f_i|$$

M20

$$M_{tot} = \sum_{i}^{n} M_i = \sum_{i}^{n} f_i\left[(x_i - x_c)^2 + (y_i - y_c)^2\right]$$

$$M_{20} = \log_{10}\left(\frac{\sum_i}{M_{tot}}\right) \text{ while } \sum i f_i < 0.2 f_{\mathrm{tot}}$$

Filamentarity

$$\mathrm{F} = 1 - \left((\text{ isophotal area })/(\pi \times (a/2)^2)\right)$$

MID

Multimode

$$g_{i,j} = \begin{cases} 1 & f_{i,j} \geqslant q_l \\ 0 & \text{otherwise} \end{cases} \qquad R_l = \frac{A_{l,(2)}}{A_{l,(1)}}$$

$$M = \max R_l$$

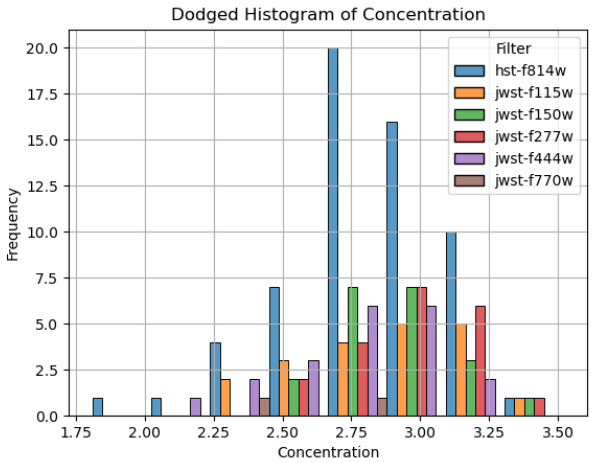
Intensity

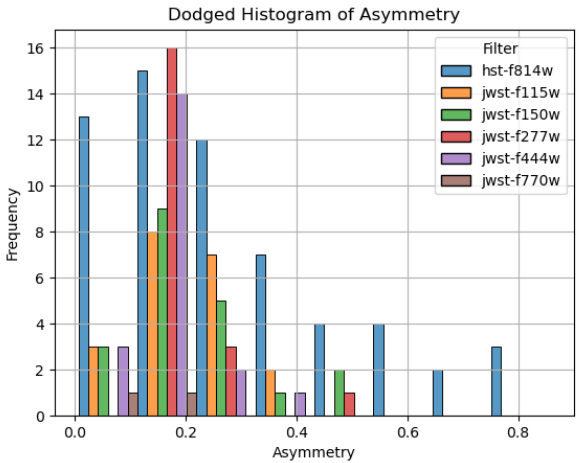
$$I = \frac{I_{(2)}}{I_{(1)}}$$

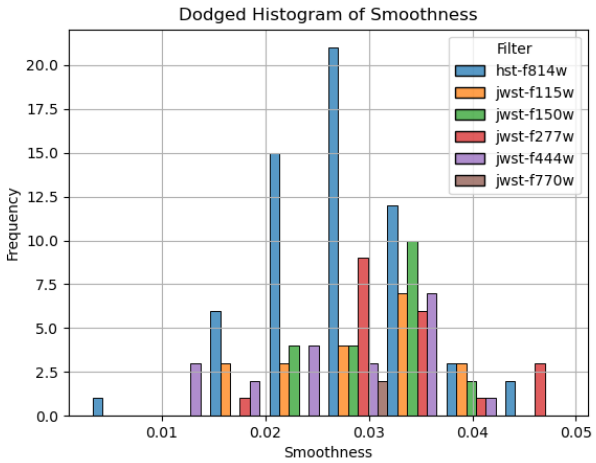
Deviation

$$D = \sqrt{\frac{\pi}{n_{\mathrm{seg}}}}\sqrt{\left(x_c - x_{l_{(1)}}\right)^2 + \left(y_c - y_{l_{(1)}}\right)^2}$$

Results CAS

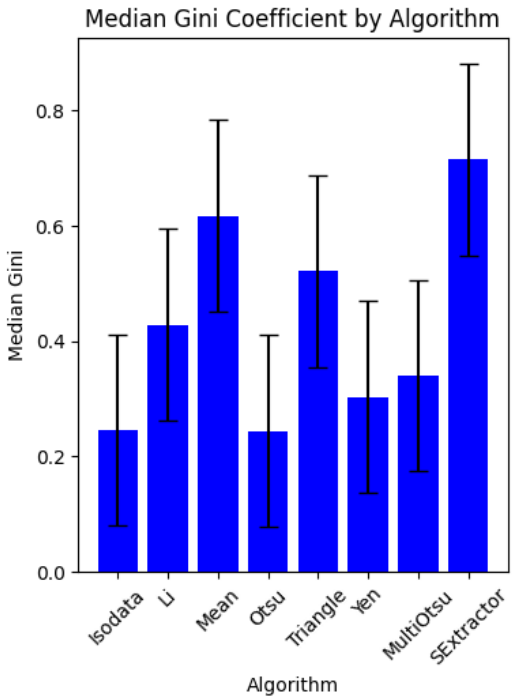

Dodged Histogram of Concentration
Frequency
Concentration
Filter
hst-f814w
jwst-f115w
jwst-f150w
jwst-f277w
jwst-f444w
jwst-f770w

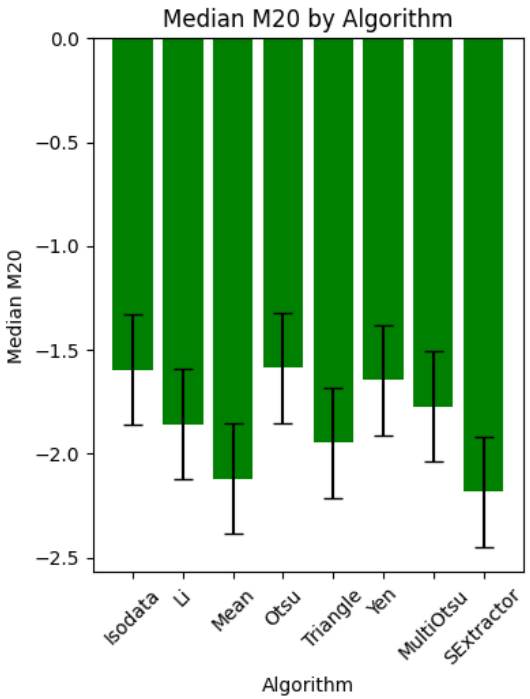

Dodged Histogram of Asymmetry
Frequency
Asymmetry
Filter
hst-f814w
jwst-f115w
jwst-f150w
jwst-f277w
jwst-f444w
jwst-f770w

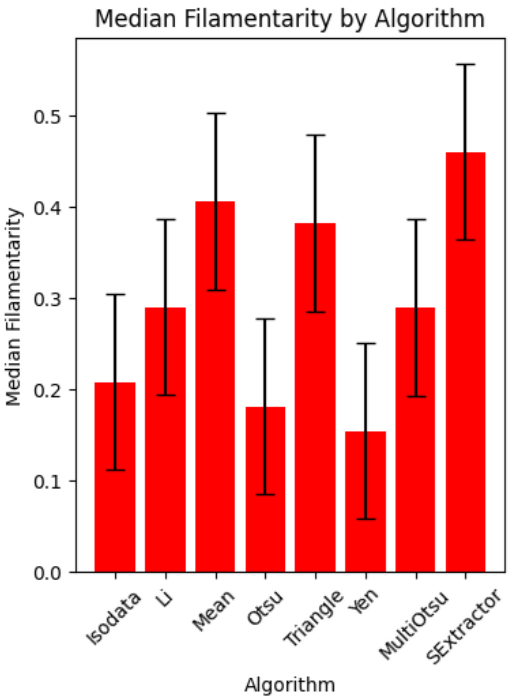

Dodged Histogram of Smoothness
Frequency
Smoothness
Filter
hst-f814w
jwst-f115w
jwst-f150w
jwst-f277w
jwst-f444w
jwst-f770w

Results Gini/M20 & Filamentarity

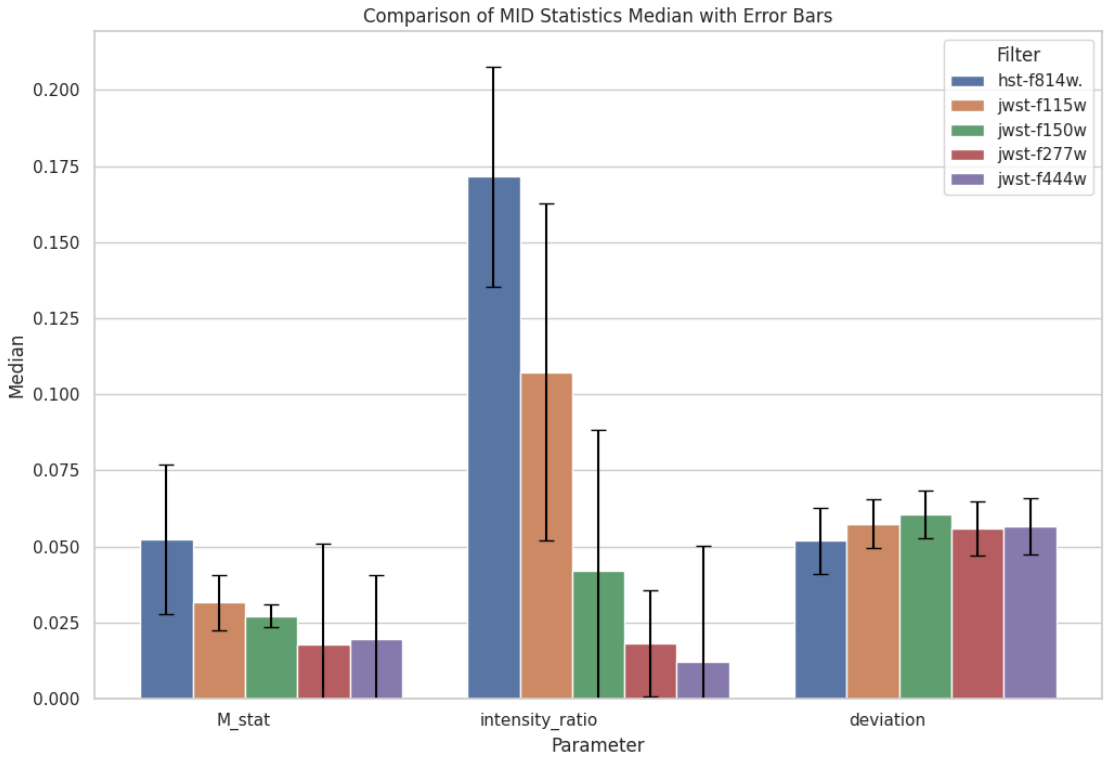
ALL.csv



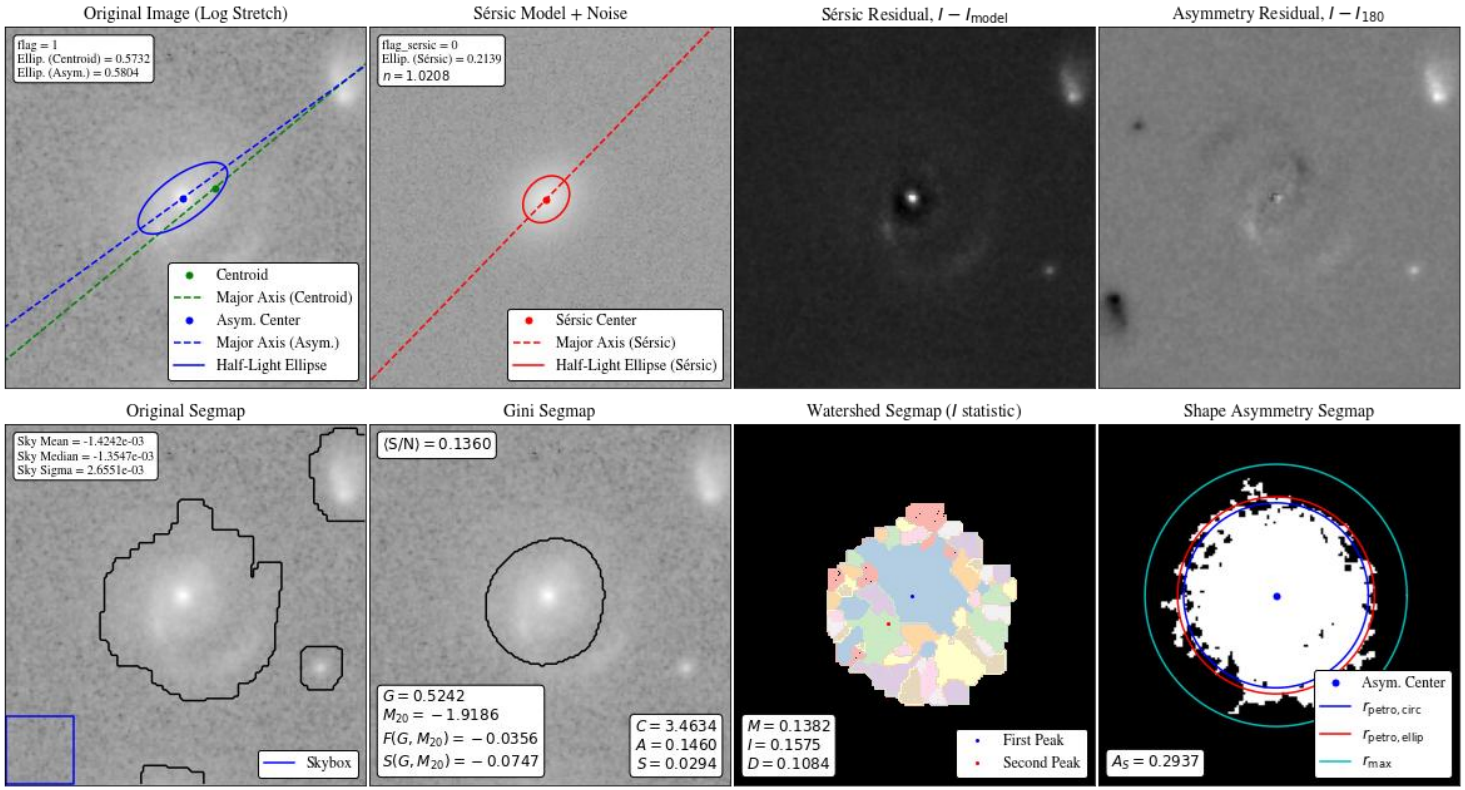




Results MID


Comparison of MID Statistics Median with Error Bars
Filter
hst-f814w.
jwst-f115w
jwst-f150w
jwst-f277w
jwst-f444w
Median
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
M_stat
intensity_ratio
deviation
Parameter

Results Galight



Discussion for the Non-Parametric Results

Based on data our findings on galaxy mergers are inconclusive. Gini and M20 results suggest mergers due to Gini/M20 sensitivity to both minor and major mergers. CAS which detects mainly major mergers, indicates no major mergers. However, wavelength-dependent concentration values suggest variability. MID statistics also point towards non-merger galaxies.

Discussion for the Parametric Results

Although the Sersic index , having the values between 3-4, we can’t conclude about the merger history of this galaxies. But one thing to note is that the Sérsic profile does not fit very well with more "disturbed" galaxies.

Most of the effective radius values indicate that the galaxies were in the process of growing and evolving but had not reached the size of larger galaxies.

The ellipticity, having very low values, indicates that the majority of galaxies were more elongated than circular, which indicates that they would probably be mergers.

Conclusion

In conclusion, we do not have results supporting any far side of the theory, some methods agree with the galaxies being mergers while others do not, however the small amount of time and data that we had available did not allow us to analyze the data with the utmost rigor, leaving the answer to the question open.

The topic is definitely interesting and worth exploring further in the near future in order to reach a answer to whether galaxy mergers make the AGN more likely to be Type II.

Acknowledgements

Project Contributors:

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- Fábio Baptista
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- Lívia Oliveira
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- Shaik Samad Pasha
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